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# Use of ML and XAI as decision–support for Business Intelligence analysis.

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# Decision suport challenges

Align business objectives with technical aspects of ML model training

Universality or specialization?

Add context for interpretation of XAI results to enhance usefulness of predictions

Challenges

Technical perspective

- Performance
- Explainability

Usability perspective

- Evaluation
- Visualisation

Inefficient model → **wrong conclusions**

Unclear results → **no conclusions**

# Usability of ML in Business Intelligence

Motivations for investment in Industry 4.0 - Company **profit**

Revenue

**Cost** optimisation

**Time** optimisation

**Quality** optimisation

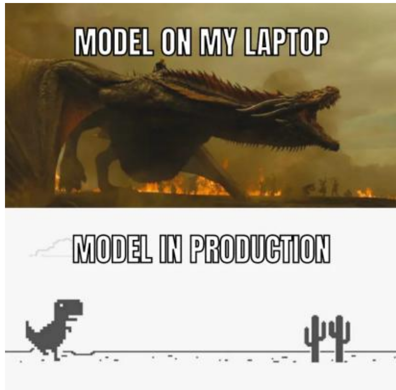
## Tasks of ML as Decision Support

Mimic current abilities

Interpretable and consistent with domain knowledge

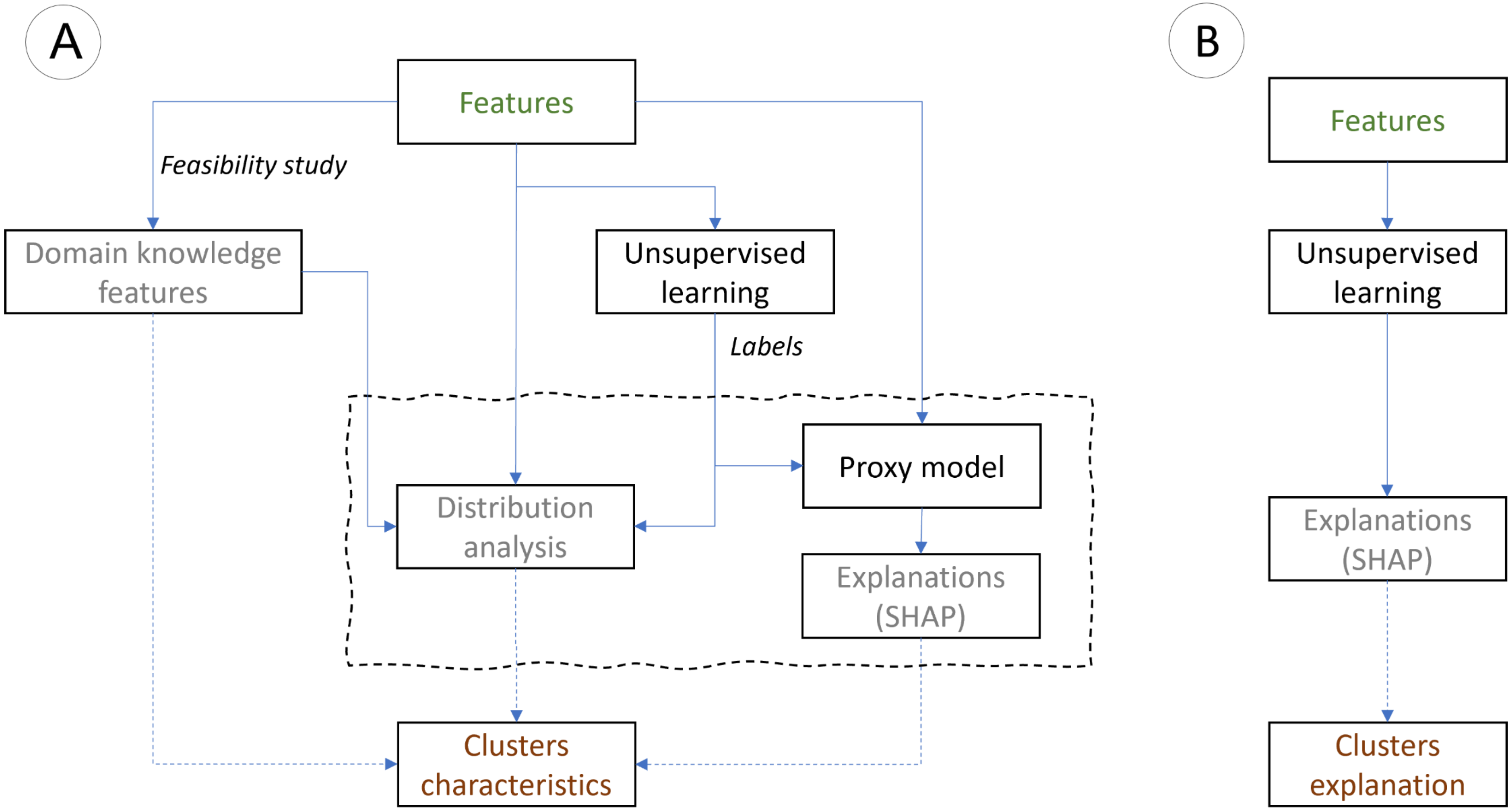
Enhance current abilities

Need of new metrics, and context for new results interpretation

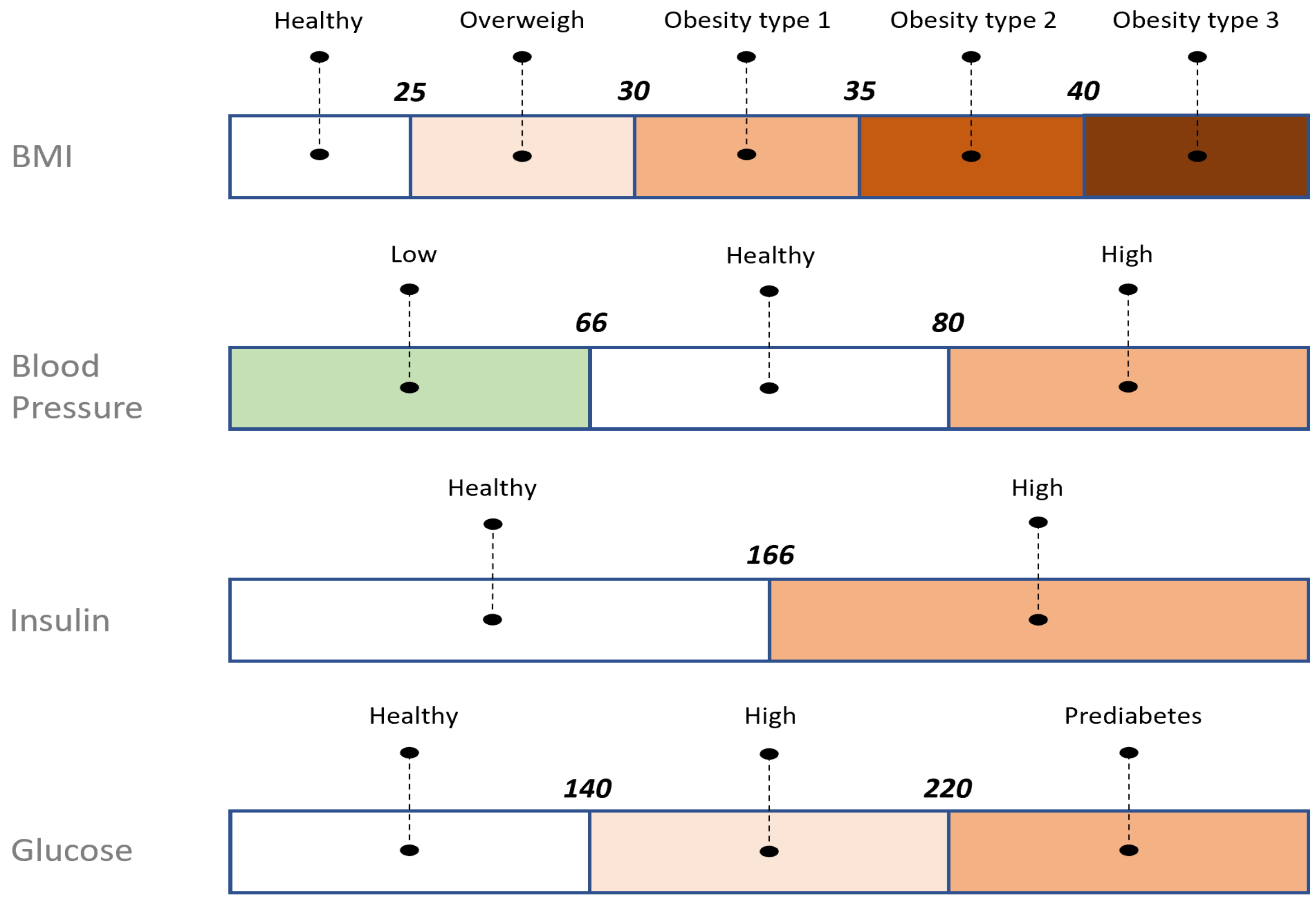


# „Improving understandability of explanations with a usage of expert knowledge – unsupervised learning”

*Maciej Szelążek, Szymon Bobek, Grzegorz J. Nalepa*



# Feasibility study of domain knowledge

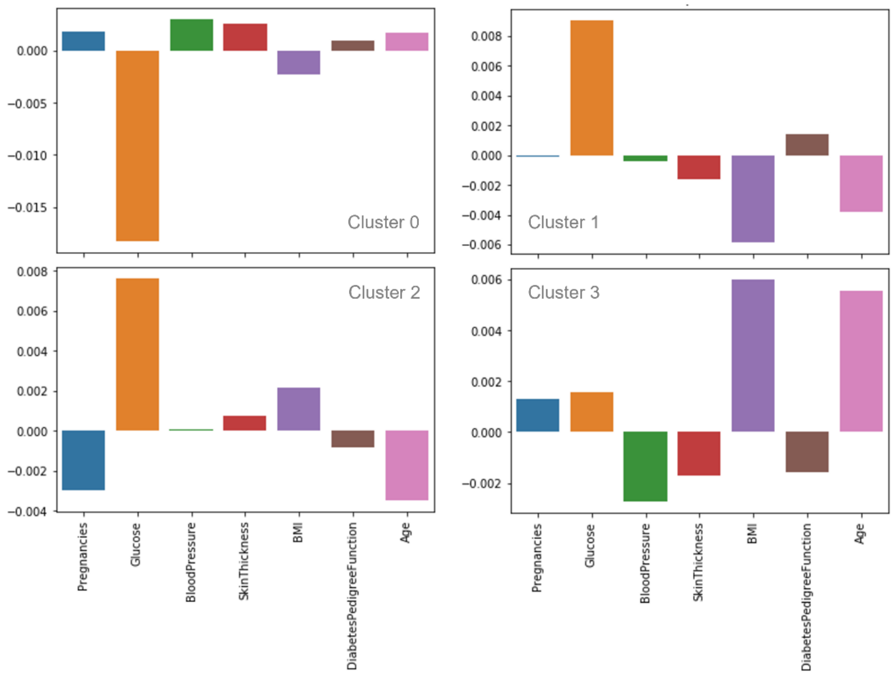


# Multiple information sources for describing clusters

Original features characteristics

| **Cluster** | Pregnancies | Glucose | BloodPressure | SkinThickness | BMI | DiabetesPedigreeFunction | Age |
|---|---|---|---|---|---|---|---|
| 0 | **5,0** | **168,7** | **76,6** | 25,1 | **35,1** | **0,55** | **38,7** |
| 1 | 3,5 | 121,7 | 73,7 | **33,6** | 34,9 | 0,50 | 31,9 |
| 2 | 4,6 | 121,9 | 75,9 | *2,3* | 30,7 | *0,42* | 37,7 |
| 3 | *2,9* | *90,8* | *66,2* | 21 | *29,8* | 0,44 | *28,2* |

SHAP characteristics




Domain features characteristics

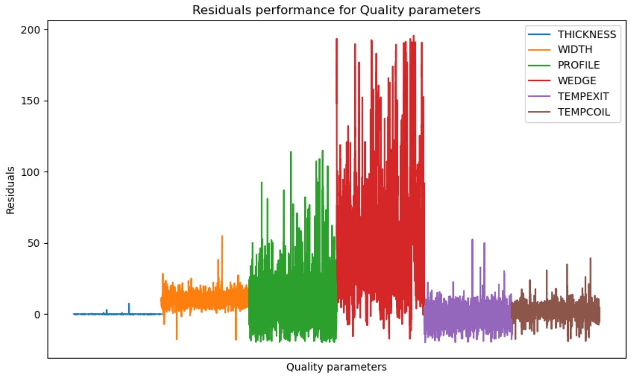
| **Cluster** | BMI_obese3 | BMI_obese2 | BMI_obese1 | BMI_overweight | Glucose_hi | BloodPressure_hi | BloodPressure_lo |
|---|---|---|---|---|---|---|---|
| 0 | 32 | 33 | 57 | 20 | **153** | **54** | 24 |
| 1 | **34** | **53** | **58** | 40 | 16 | 51 | 46 |
| 2 | 12 | 18 | 42 | 42 | 16 | 40 | 25 |
| 3 | 12 | 37 | 55 | **64** | 0 | 18 | **111** |

# „Use of model explainability as actionable Decision–Support for data series analysis"

*Maciej Szelążek, Szymon Bobek, Grzegorz J. Nalepa*

## Goal: Add context for interpretation of XAI results

- Contextualisation - Presentation of XAI in the context of quality indicators consistent with the current methodology of quality management in the organization
- Actionability
    - Automation of the procedure
    - Universal KPIs - The procedure allows for any dependent variable (quality metric), regardless of the input data
- Visualisation
    - Present the results in a way that can be understood by a diverse range of audiences



# KPIs - Capability factors

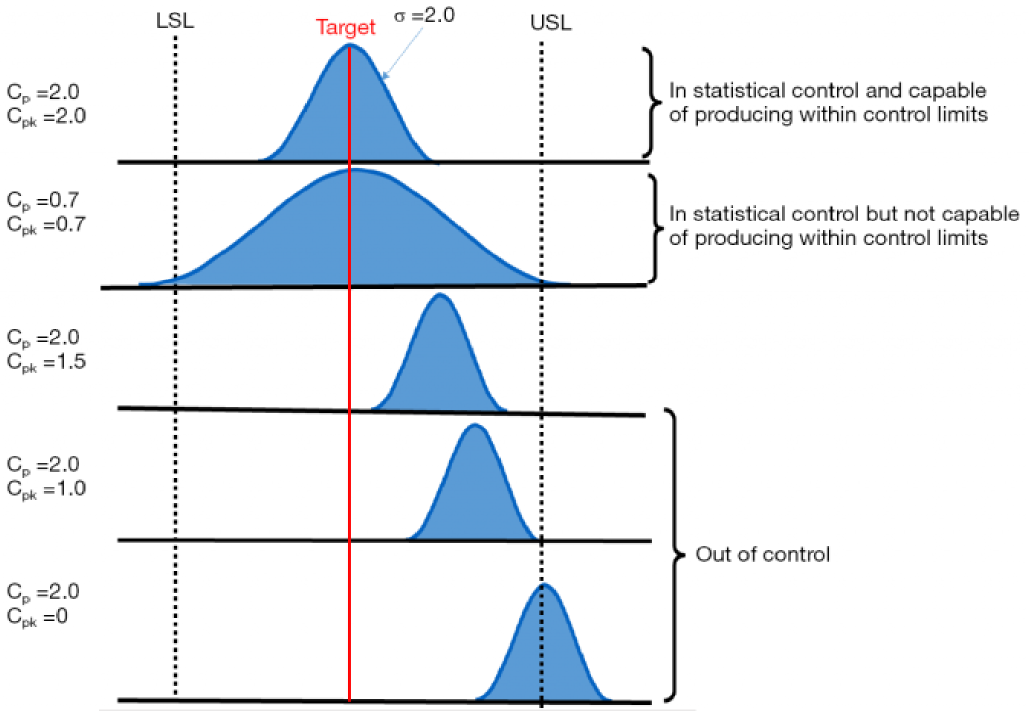
**Process capability – Cp :** tolerance width divided by the total spread of process (6 Sigma).

**Calculation of Process Capability (Cp) :**

$$\mathbf{Cp} = \frac{\text{Design Tolerance}}{6\sigma} = \frac{\text{USL - LSL}}{6\sigma}$$

**USL** = Upper Specification Limit, **LSL** = Lower Specification Limit

| Cp | Defects amount |
|---|---|
| 1 | 2700 ppm |
| 1,33 | 63 ppm |
| 1,67 | 0,57 ppm |
| 2 | 0,002 ppm |




**Process Capability Index – Cpk**: indicates shifting of the process, the minimum of Cpk upper and Cpk lower.

**Calculation of Process Capability Index (Cpk) :**

$$Cpk_U = \frac{USL - \bar{\bar{X}}}{3\sigma} \quad \text{and} \quad Cpk_L = \frac{\bar{\bar{X}} - LSL}{3\sigma}$$

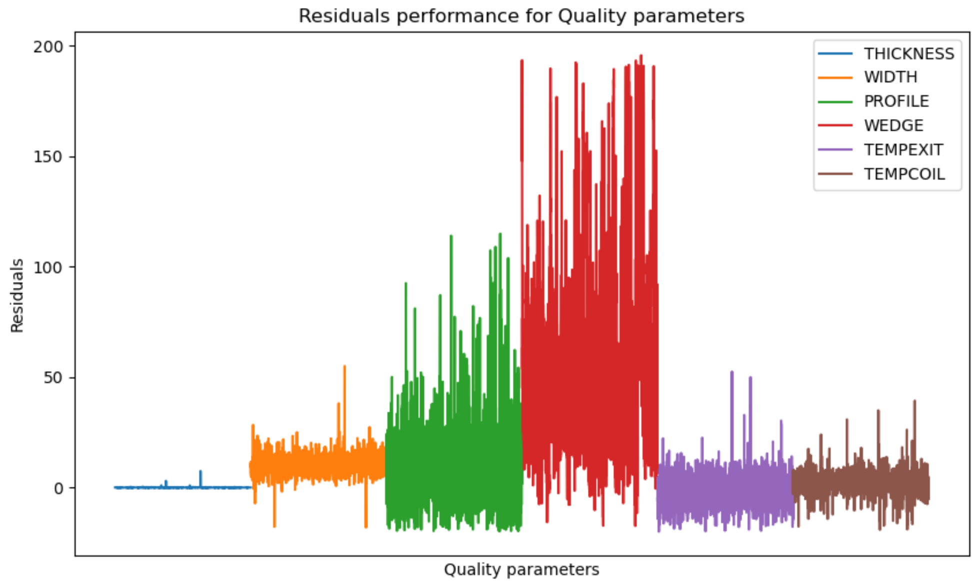
https://www.doriane-copar.com/process-capability-index-cpk/

https://techqualitypedia.com/cp-and-cpk/

# Issue – how to interpret XAI results

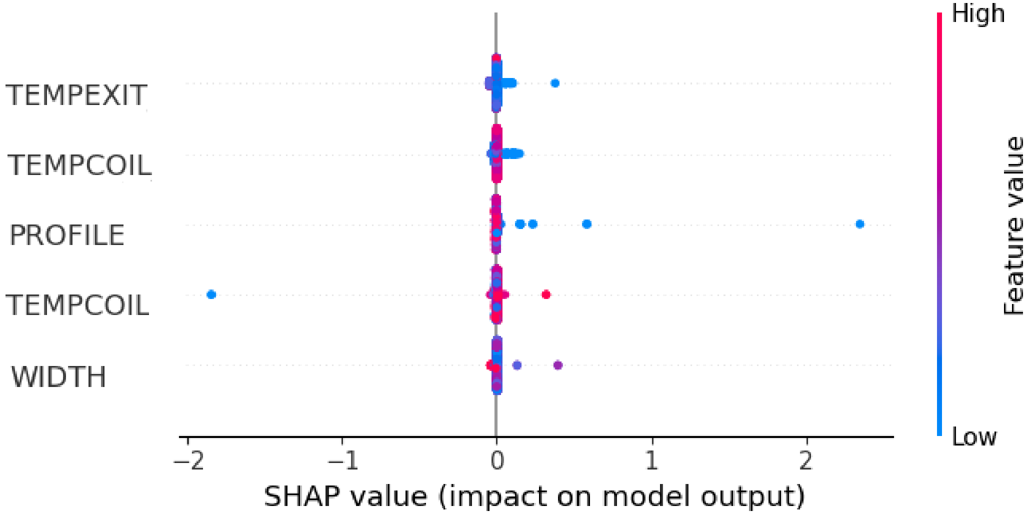
Lots of information based on analysis of KPI residuals:

- Different levels of overruns – amplitude
- Trends above and below target – density
- Cases significantly different from natural variability - outliers

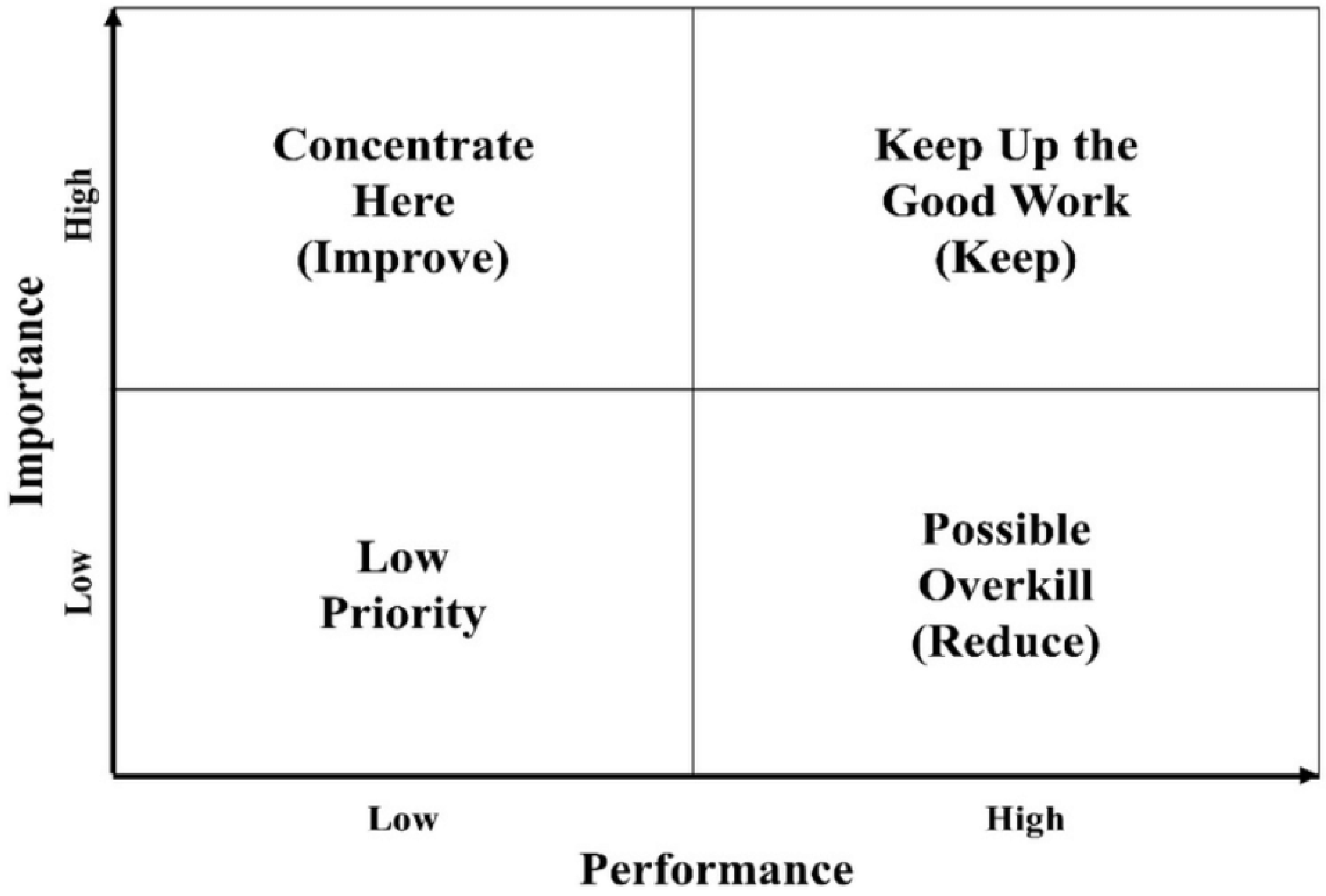



SHAP

- Similar XAI values for extremely different feature values
- No clear relationships between XAI values for different features



# Importance Performance Analysis

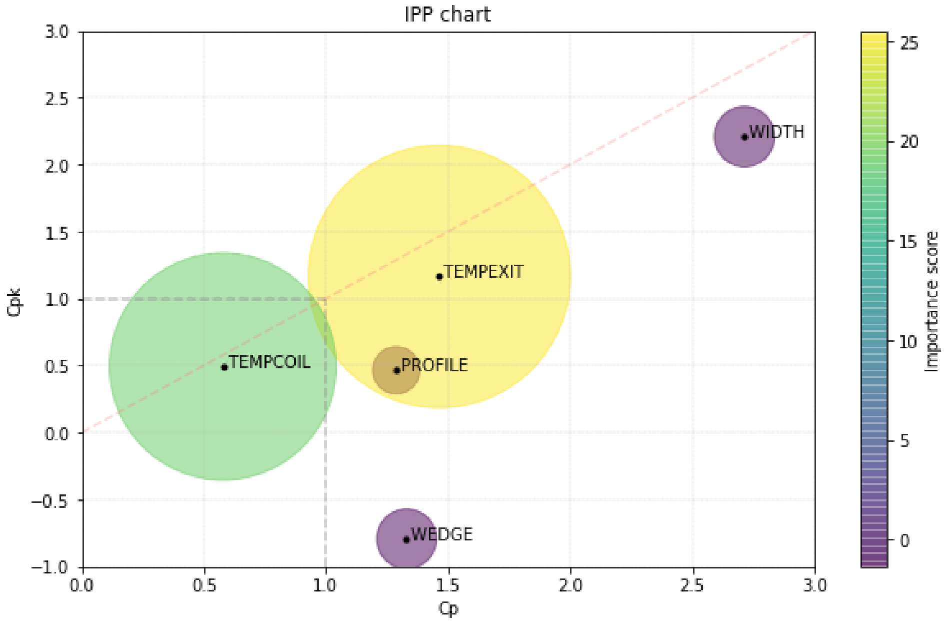


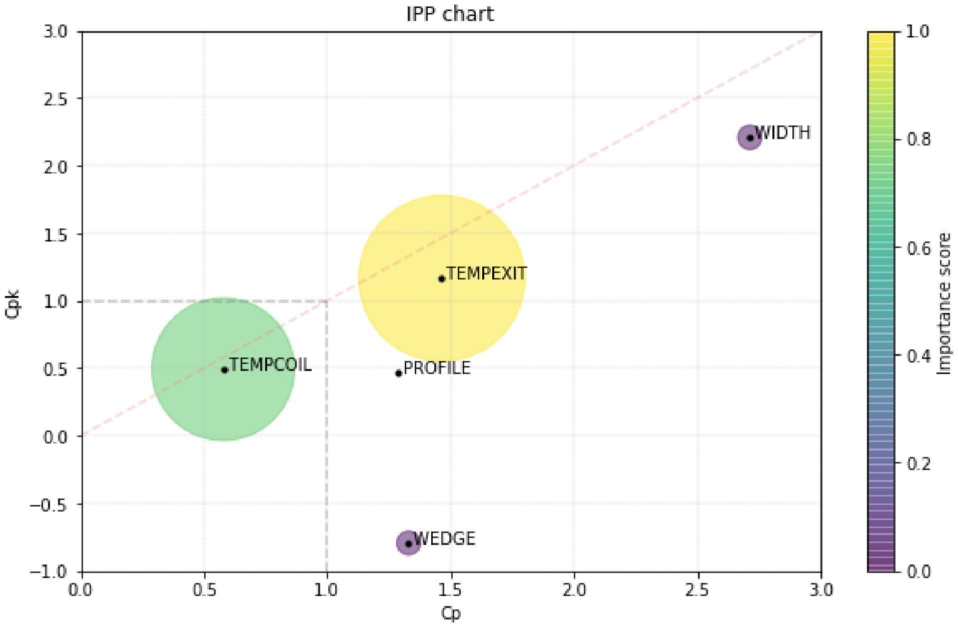
# Results - IPP chart

The closer to the **red line** the more centered the process

The further outside the gray lines the better.

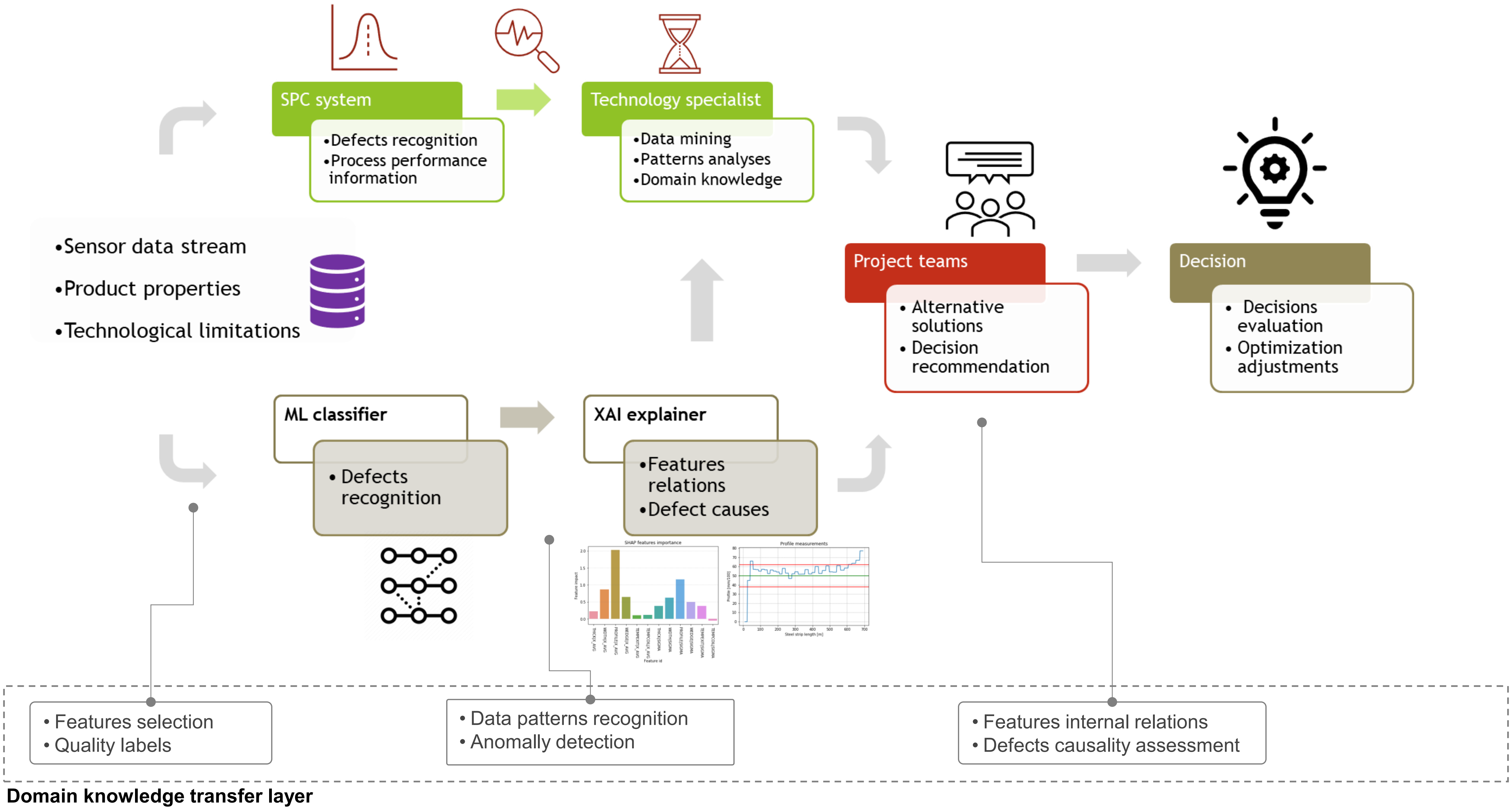
The table shows the sign of the XAI score, and the precise scale of differences between the features importances.




| Feature | Cp | Cpk | residuals shap |
|---|---|---|---|
| WIDTH | 2.71 | 2.21 | -1.35 |
| PROFILE | 1.29 | 0.46 | -0.83 |
| WEDGE | 1.33 | -0.79 | -1.33 |
| TEMPEXIT | 1.46 | 1.16 | 25.54 |
| TEMPCOIL | 0.58 | 0.49 | 19.10 |

| Feature | Cp | Cpk | residuals shap |
|---|---|---|---|
| WIDTH | 2.71 | 2.21 | 0.02 |
| PROFILE | 1.29 | 0.46 | 0.00 |
| WEDGE | 1.33 | -0.79 | 0.02 |
| TEMPEXIT | 1.46 | 1.16 | 1.00 |
| TEMPCOIL | 0.58 | 0.49 | 0.74 |


SPC system
•Defects recognition
•Process performance information
Technology specialist
•Data mining
•Patterns analyses
•Domain knowledge
Project teams
• Alternative solutions
• Decision recommendation
Decision
• Decisions evaluation
• Optimization adjustments
•Sensor data stream
•Product properties
•Technological limitations
ML classifier
• Defects recognition
XAI explainer
•Features relations
•Defect causes
SHAP features importance
Profile measurements
• Features selection
• Quality labels
• Data patterns recognition
• Anomally detection
• Features internal relations
• Defects causality assessment
Domain knowledge transfer layer

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Thank you for your attention